Report_DEB

1. Importation des données

```
file_path <- here::here("mod/DEB_SASWI")

l_Experiments <- c(
  "Bart2019_XP1_1",
  "Bart2019_XP1_2",
  #"Bart2019_XP2",
  "Bart2019_XP3",
  "Bart2019_XP4_1",
  "Bart2019_XP4_2",
  "Bart2019_XP5",
  "Bart2020",
  #"Gollot2026_lufa"
  "Gollot2026_dens_D1",
  "Gollot2026_dens_D2",
  "Gollot2026_dens_D2b",
  "Gollot2026_dens_D5",
  "Gollot2026_dens_D7"
  )

Adults_alone <- FALSE

df_data <- f_import_data_DEB(l_Experiments, Adults_alone)
```

2. Model fit

```r
NbIter <- 100000
#seeds <- c(C1 = 3336, C2 = 6663, C3 = 1222, C4 = 2420, C5 = 4848, C6 = 5656, C7 = 7077, C8 =
seeds <- c(C1 = 3333, C2 = 6666, C3 = 1222)

text_priors <- '
Distrib (pAm,          TruncNormal_cv,        100,      0.2,    40, 2000);
Distrib (Fm,           TruncNormal,            0.1,      0.2,    0.01, 0.8);
Distrib (r_pAm_pM,     TruncNormal_cv,         0.8,      0.1,    0.1, 2);
Distrib (kap,          Uniform,                0.1,     0.9);
Distrib (Ehp,          TruncNormal_cv,         100,      0.2,    40, 300);
Distrib (E_coc,        TruncNormal_cv,              30,      0.2, 1, 300);
Distrib (rOM_ClxHorse,  LogUniform,           0.00001,      0.3);
Distrib (f_Slope, TruncNormal_cv, 1, 1, 0, 10);
Distrib (dv, TruncNormal_cv, 1, 0.2, 0, 5);
Distrib (w, TruncNormal_cv, 20, 10, 0, 50);

# Calcul des vraissemblances
Distrib (Sigma_W, TruncNormal,  5, 1, 0.01, 100);'

text_likelihood <- 'Likelihood (Weight,       Normal, Prediction(Weight), Sigma_W);
Likelihood (Reproduction, Poisson, Prediction(Reproduction));'

OM_diff <- TRUE

# makemcsim DEBcali.model

f_In_tot(file_path, l_Experiments, Adults_alone, OM_diff, text_priors, text_likelihood, NbIte

# ./mcsim.DEBcaliSWI DEB_C1.in
```

`OM_horse` and `OM_soil` distinct ? TRUE

Adult and alone earthworm taken into account ? FALSE

3. Results

```
               pAm.1.     Fm.1. r_pAm_pM.1.      kap.1.    Ehp.1.  E_coc.1.
Result_mode 174.45000 0.5568970  0.86177900 0.61565400  95.36520  6.898790
2.5%        145.06585 0.3390000  0.73819800 0.59042505  80.06250  6.038910
97.5%       200.89900 0.7634520  0.96419400 0.64595745 129.48085 10.056940
Min          83.18490 0.0410025  0.67703500 0.27393000  63.11510  5.063920
```

```
Max         213.47900 0.7999790  1.01783000 0.66497400 142.68900 30.709500
Result_mean 173.35245 0.5415368  0.84759216 0.61861362 103.43182  7.590457
Result_sd    13.64922 0.1083505  0.05481412 0.01437679  12.36677  1.000422
            rOM_ClxHorse.1. f_Slope.1.      dv.1.      w.1. Sigma_W.1.
Result_mode    0.0012002100 0.29041700 1.0148100 48.433200 0.12524300
2.5%           0.0007084183 0.25424100 0.6345946 29.616545 0.11115600
97.5%          0.0043149800 0.41147700 1.4139400 49.689900 0.14559900
Min            0.0000340354 0.22235400 0.2535420 12.268800 0.09947490
Max            0.0509530000 2.06027000 1.7845100 49.999600 7.33650000
Result_mean    0.0021396054 0.32541216 1.0222531 43.030537 0.12833137
Result_sd      0.0009774100 0.04253901 0.2014200  5.363403 0.06014009

Number of observations, n =  277
Number of parameters, k =  10
Log-likelihood, LL =  -22.25737
BIC =  100.7549



Potential scale reduction factors:

                Point est. Upper C.I.
dv.1.                 1.00       1.00
E_coc.1.              1.57       2.58
Ehp.1.                1.35       1.92
f_Slope.1.            1.18       1.52
Fm.1.                 1.14       1.42
kap.1.                1.02       1.04
pAm.1.                1.03       1.07
r_pAm_pM.1.           1.04       1.07
rOM_ClxHorse.1.       1.02       1.08
Sigma_W.1.            1.02       1.07
w.1.                  1.61       2.96

Multivariate psrf

1.5
```

```r
mcmc_list_corr <- mcmc.list(
  lapply(mcmc_list, function(x) {
    mcmc(as.matrix(x)[-1, , drop = FALSE])  # <- reconvertir en mcmc
  })
```

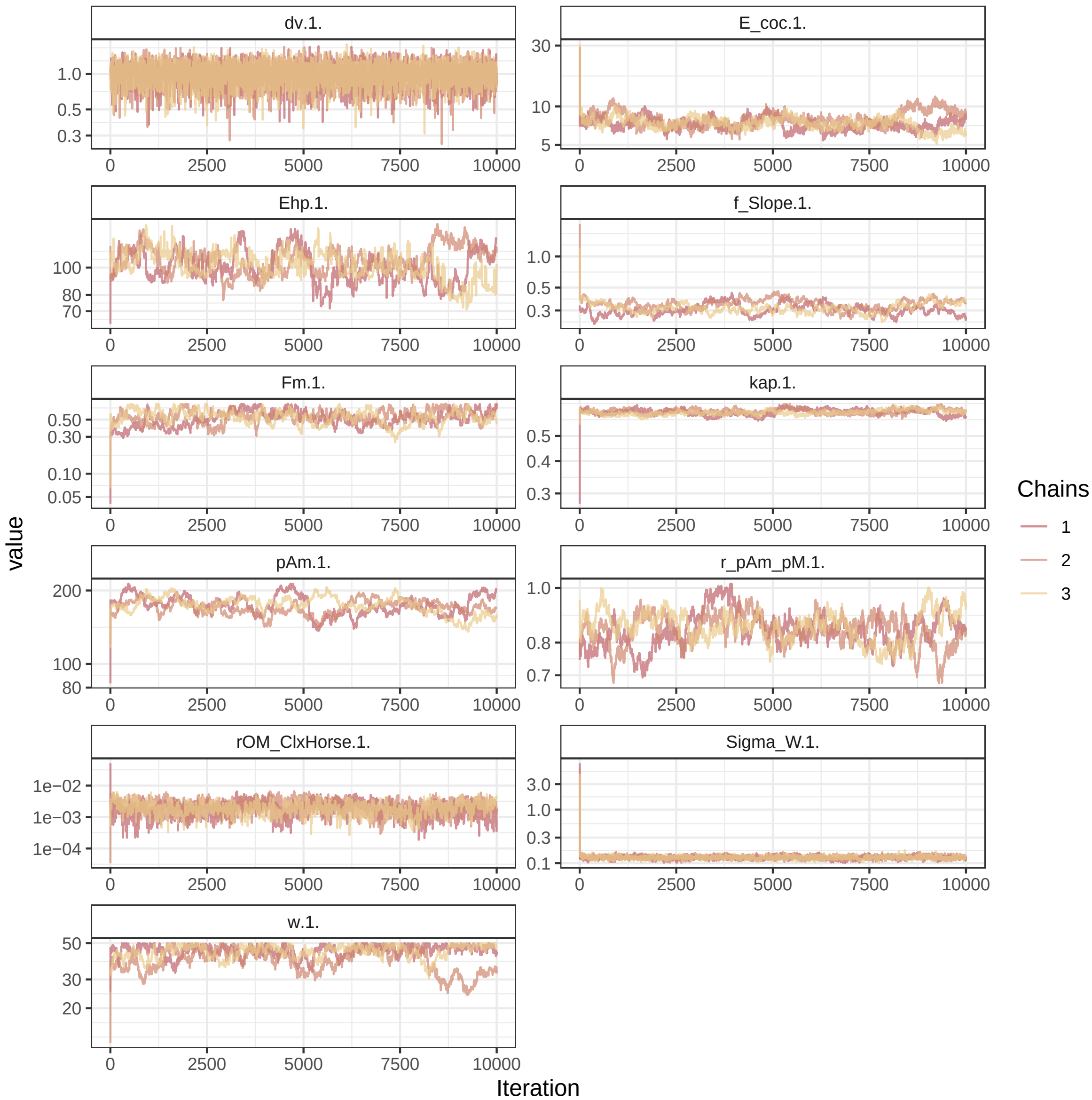


Figure 1: Parameters chains. Only the first iteration and the last `r Nb_iter_kept` are represented.

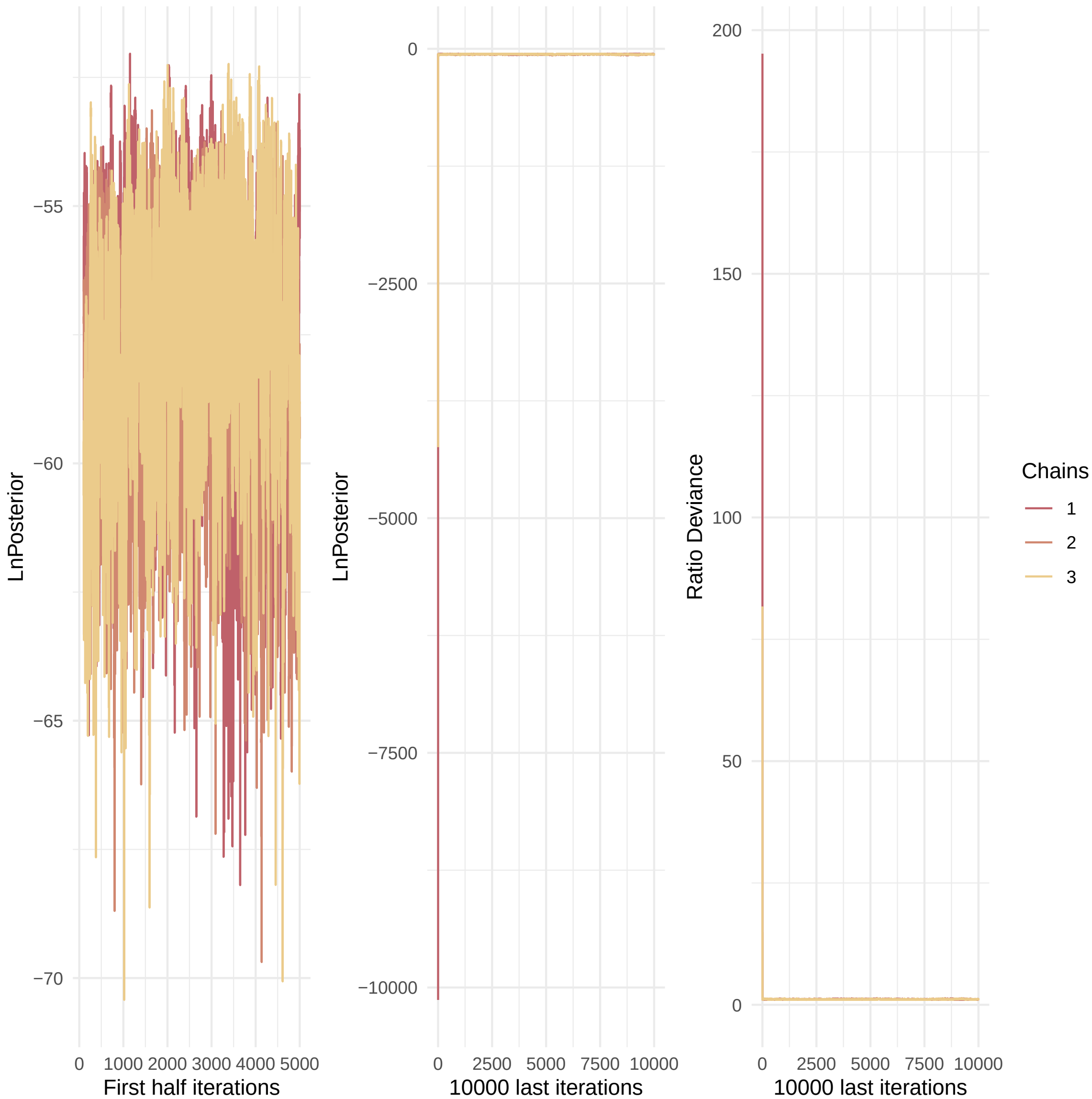


Figure 2: Evolution of likelihood and deviance ratio through the iterations.

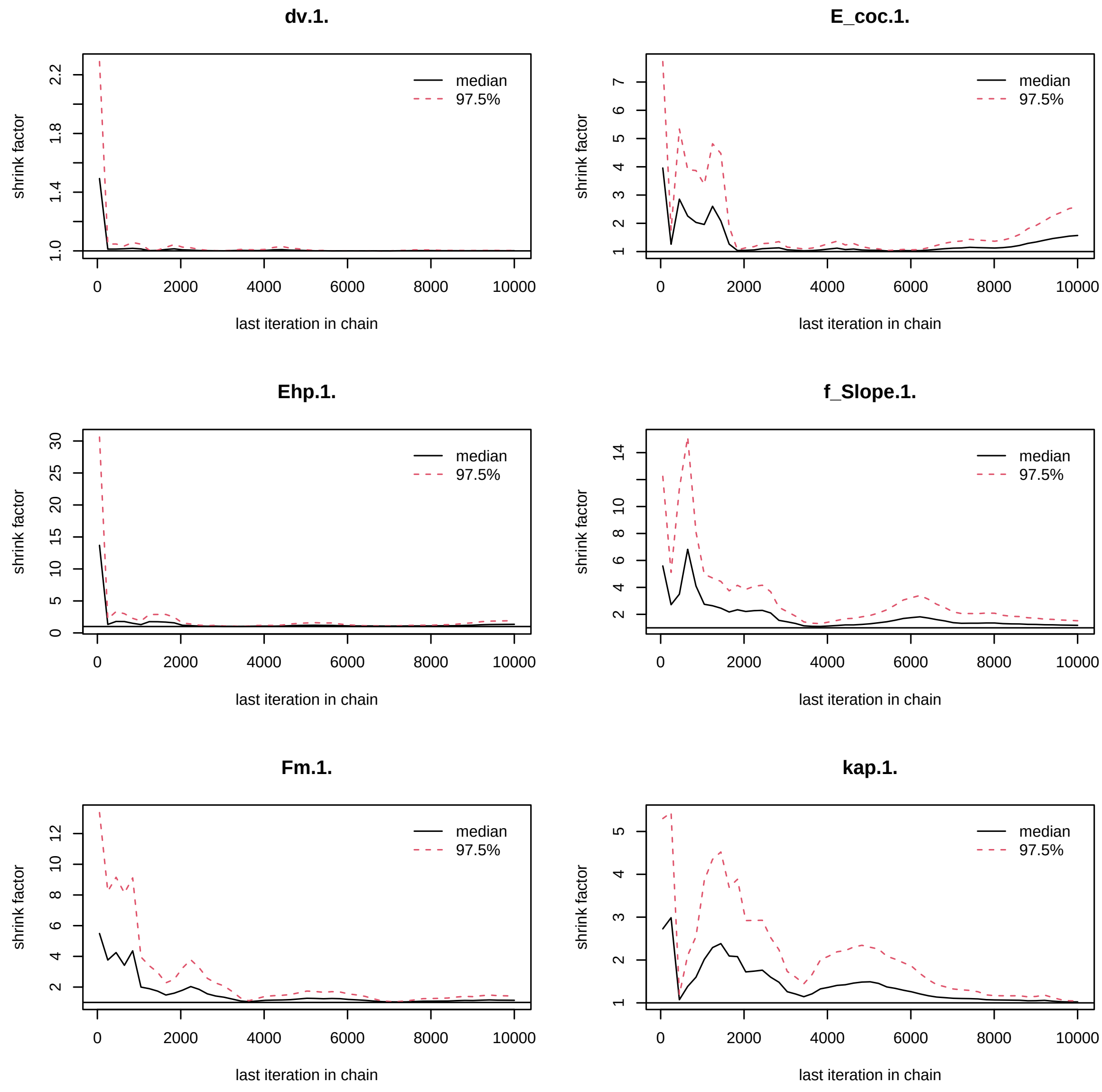


Figure 3: Chains convergence.

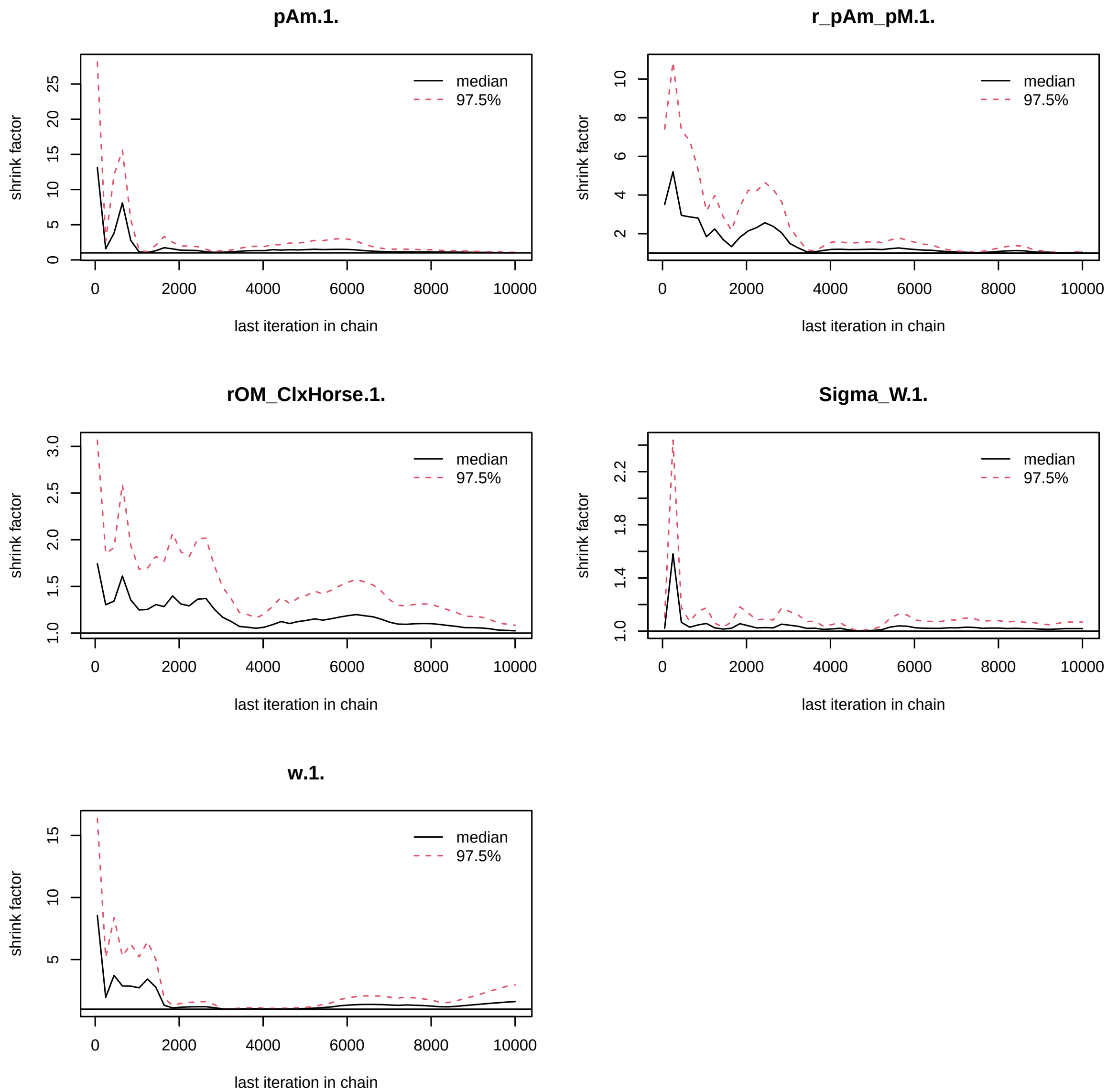


Figure 4: Chains convergence.

```
)

colramp_aurora <- colorRampPalette(rev(Nord_aurora[1:4]))
ipairs(as.matrix(mcmc_list_corr), colramp = colramp_aurora, pixs=0.25, cex.diag = 0.5)
```

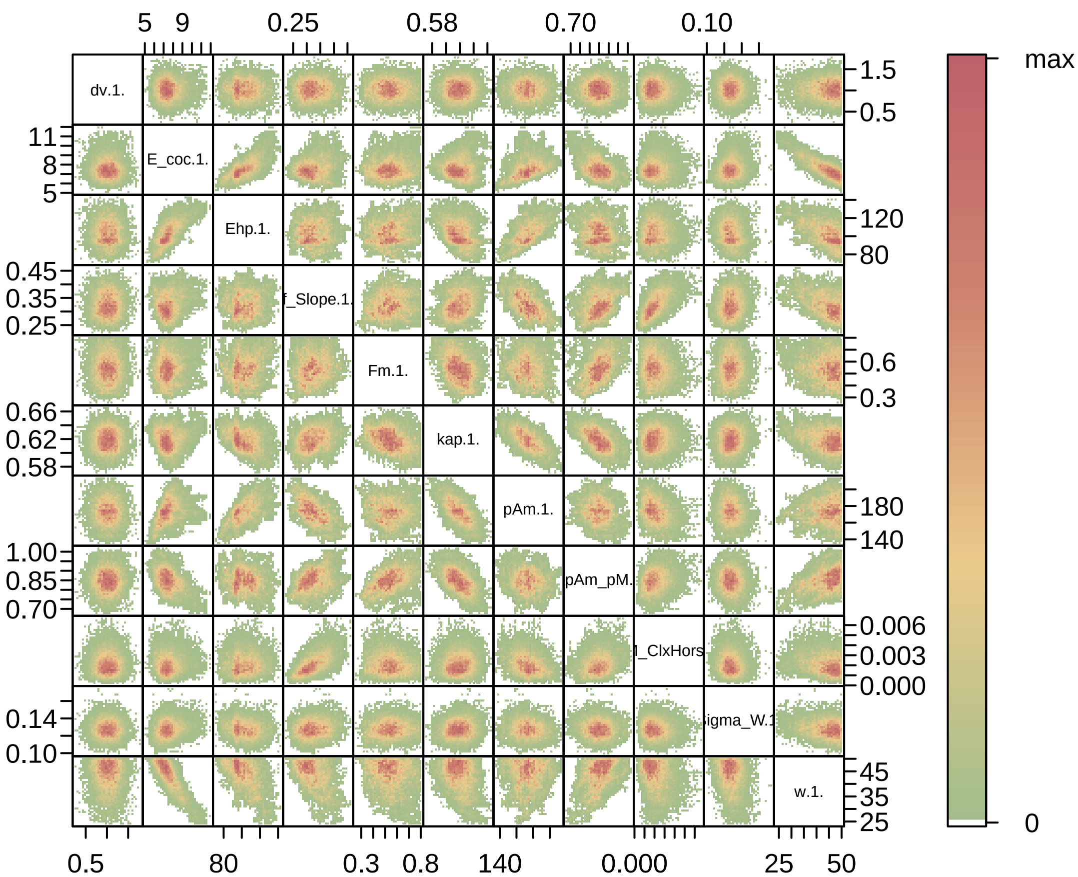


Figure 5: Parameter correlations.

4. Simulations

```
l_Experiments <- c(
  "Bart2019_XP1_1",
```

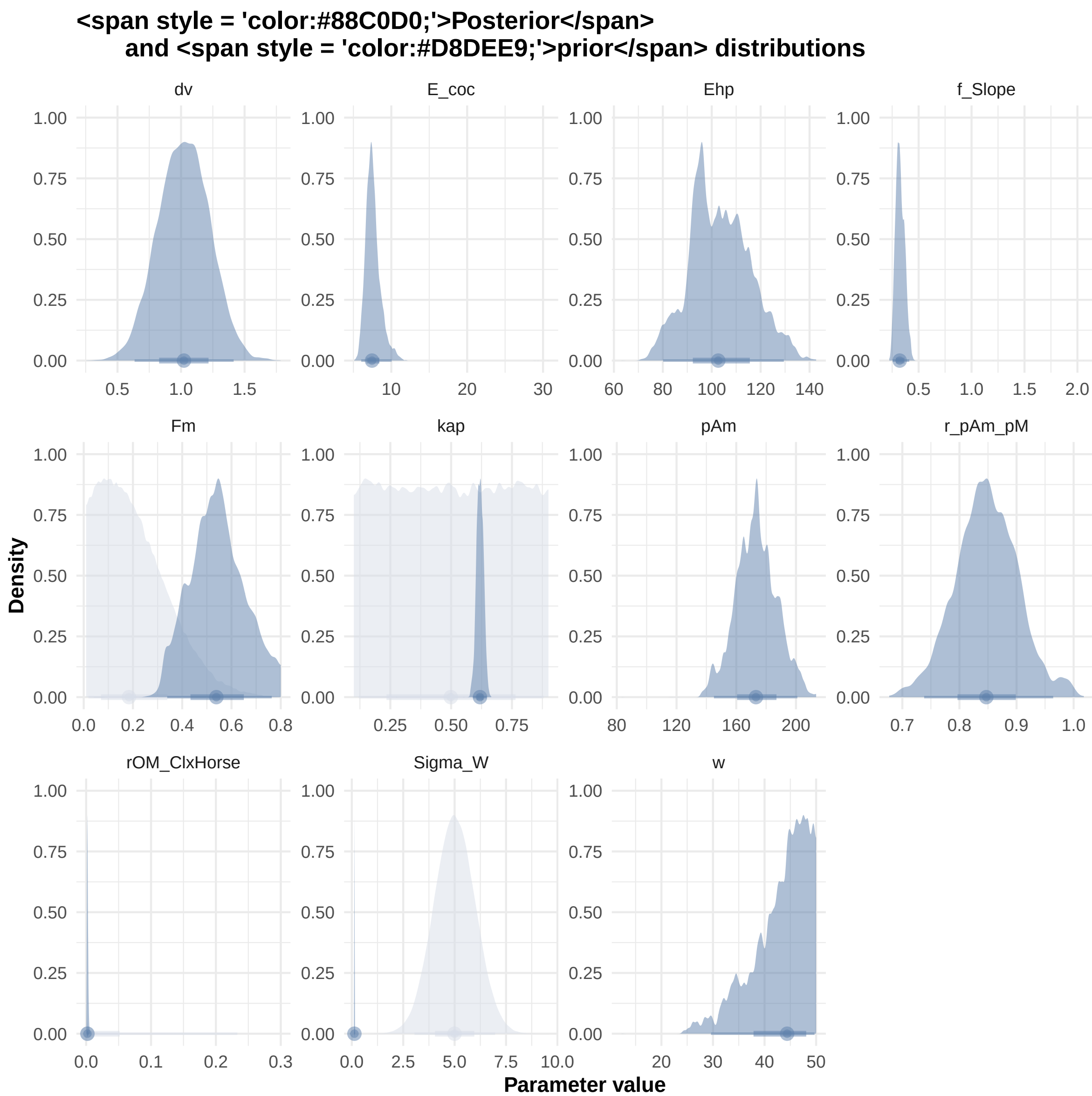


Figure 6: Distributions of the priors and the posteriors.

```r
  "Bart2019_XP1_2",
  #"Bart2019_XP2",
  "Bart2019_XP3",
  "Bart2019_XP4_1",
  "Bart2019_XP4_2",
  "Bart2019_XP5",
  "Bart2020",
  #"Gollot2026_lufa"
  "Gollot2026_dens_D1",
  "Gollot2026_dens_D2",
  "Gollot2026_dens_D2b",
  "Gollot2026_dens_D5",
  "Gollot2026_dens_D7"
  )

Adults_alone <- FALSE

df_data <- f_import_data_DEB(l_Experiments, Adults_alone)

text_param <- Summary_res["Result_mode", , drop = FALSE] %>%
  t() %>%
  as.data.frame() %>%
  tibble::rownames_to_column("param") %>%
  rename(val = 2) %>%
  mutate(
    param = gsub("\\.1\\.$|\\.1\\.", "", param),
    line  = glue("{param} = {val};")
  ) %>%
  pull(line) %>%
  paste(collapse = "\n")

Endpoints_print <- "Weight,
    Length_struct,
    Reproduction,
    Energy,
    Organic_matter,"

f_In_simulations(file_path, l_Experiments, Adults_alone, OM_diff, text_param, Endpoints_prin
#f_In_simulations(file_path, l_Experiments, Adults_alone, OM_diff=FALSE, text_param, text_pr

# ./mcsim.DEBcaliSWI DEB_sim_full.in
```

```
##############################
df_No_sim <- df_data |>
  dplyr::select(ID_experiment, No_sim)

df_carac_color <- data.frame(
  carac = unique(l_carac),
  color = c(Nord_aurora[1], Nord_polar[2], Nord_aurora[3], Nord_aurora[2], Nord_aurora[5],
            Nord_aurora[5], Nord_frost[4], Nord_frost[3], Nord_frost[2], Nord_frost[1])
)
v_color_carac <- setNames(df_carac_color$color, df_carac_color$carac)
#############################

df_data <- f_import_data_DEB(l_Experiments, Adults_alone) |>
  left_join(df_carc_exp, by="ID_experiment")

# time_limits <- df_data %>%
#   group_by(ID_experiment) %>%
#   summarise(time_cutoff = max(Time, na.rm = TRUE) + 5)

select_times <- seq(0,400, 0.1)

df_sim <- f_MCSim_read_sim(here::here(file_path, "sim.out")) |>
  left_join(df_No_sim, by="No_sim") |>
  left_join(df_carc_exp, by="ID_experiment") |>
  # left_join(time_limits, by = "ID_experiment") |>
  # filter(Time <= time_cutoff) |>
  # dplyr::select(-time_cutoff) %>%
  filter(Time %in% select_times)
```

```
Warning: `read_table2()` was deprecated in readr 2.0.0.
i Please use `read_table()` instead.
```

```
Warning in left_join(f_MCSim_read_sim(here::here(file_path, "sim.out")), : Detected an unexpe
i Row 1 of `x` matches multiple rows in `y`.
i Row 1 of `y` matches multiple rows in `x`.
i If a many-to-many relationship is expected, set `relationship =
  "many-to-many"` to silence this warning.
```

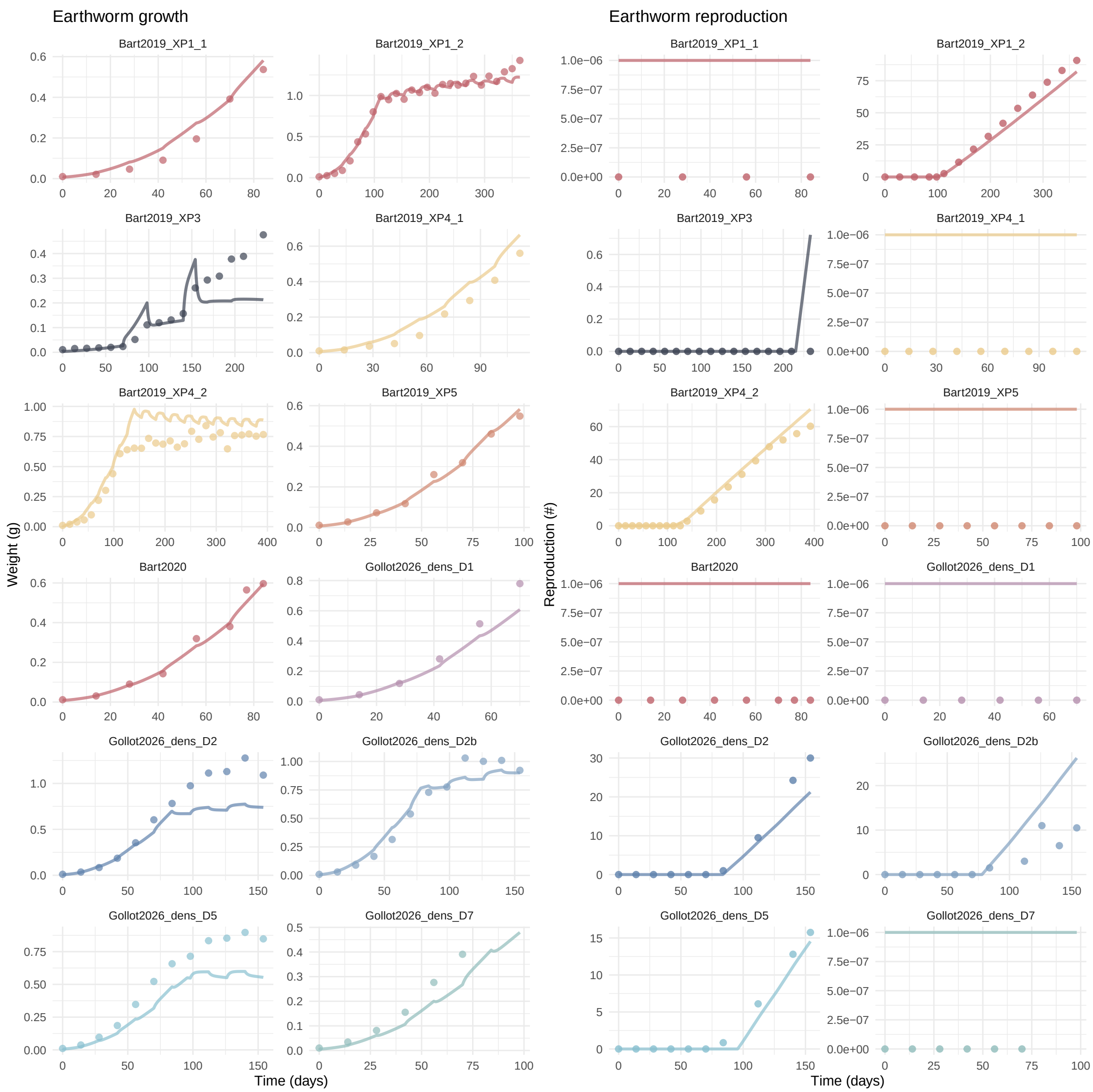
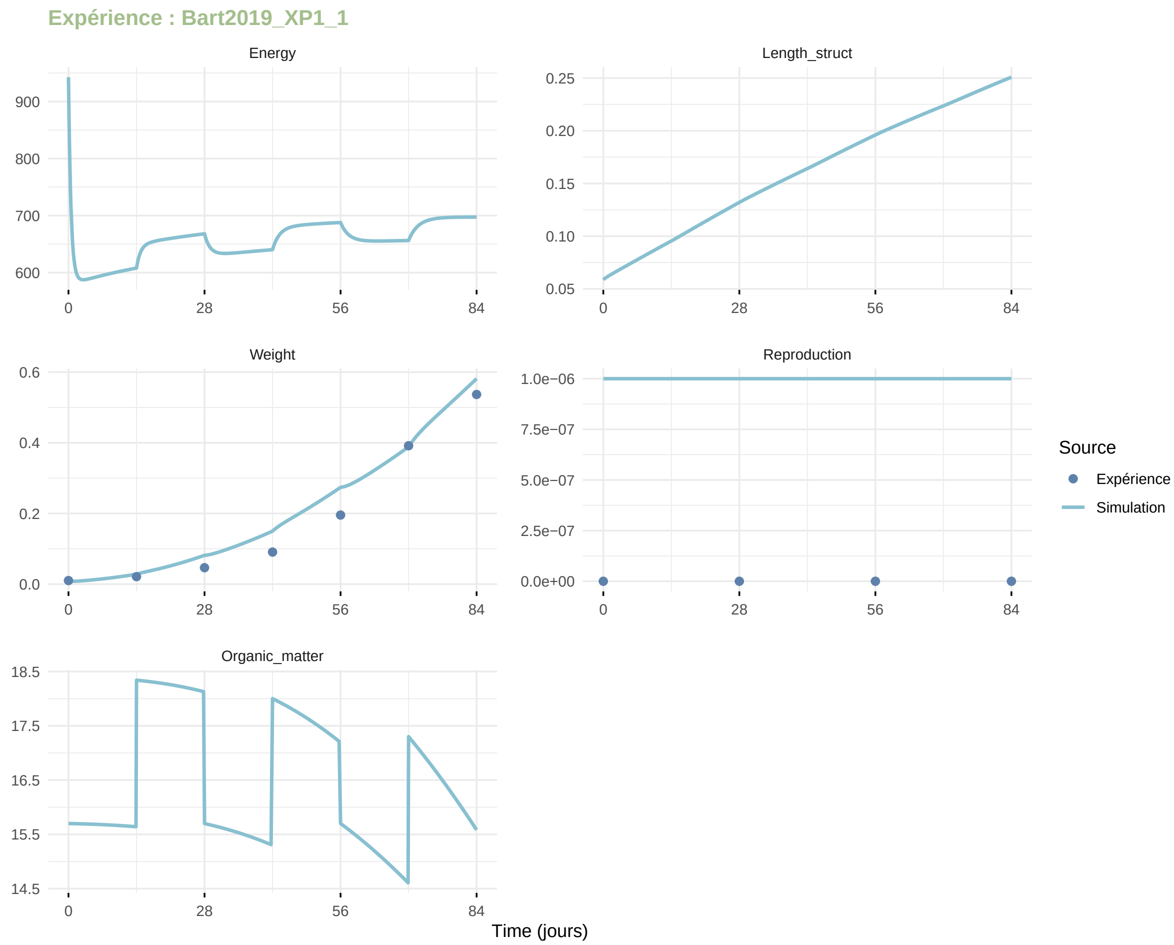
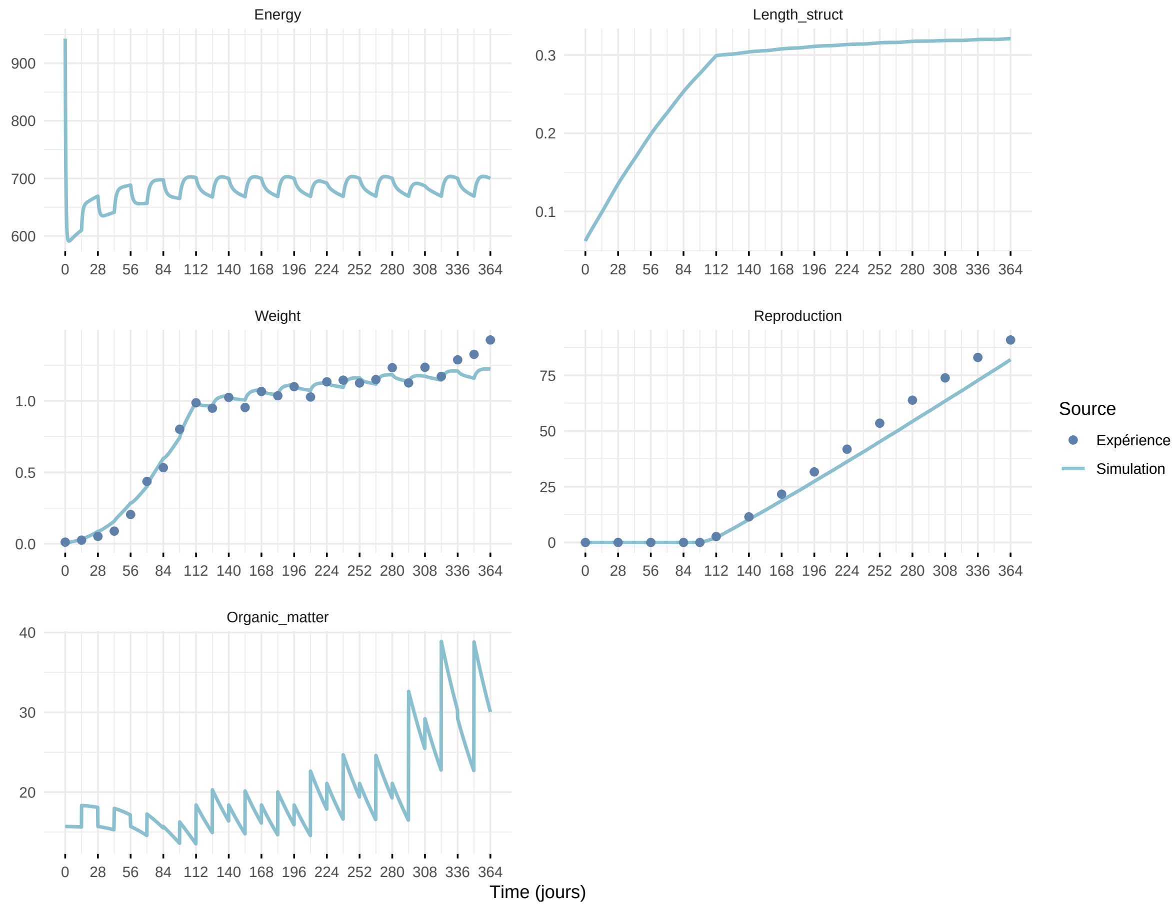


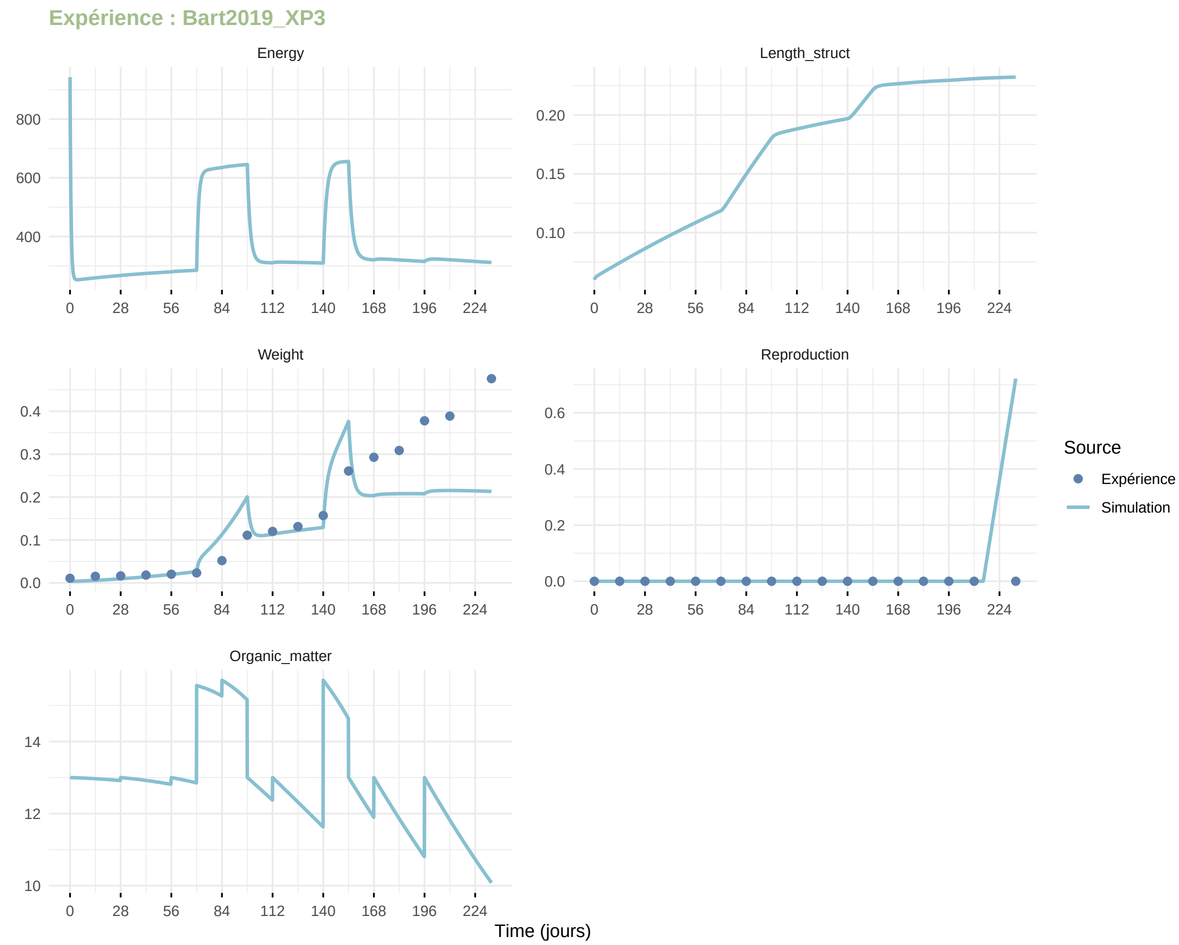
Figure 7: Simulations Weight and Reproduction.



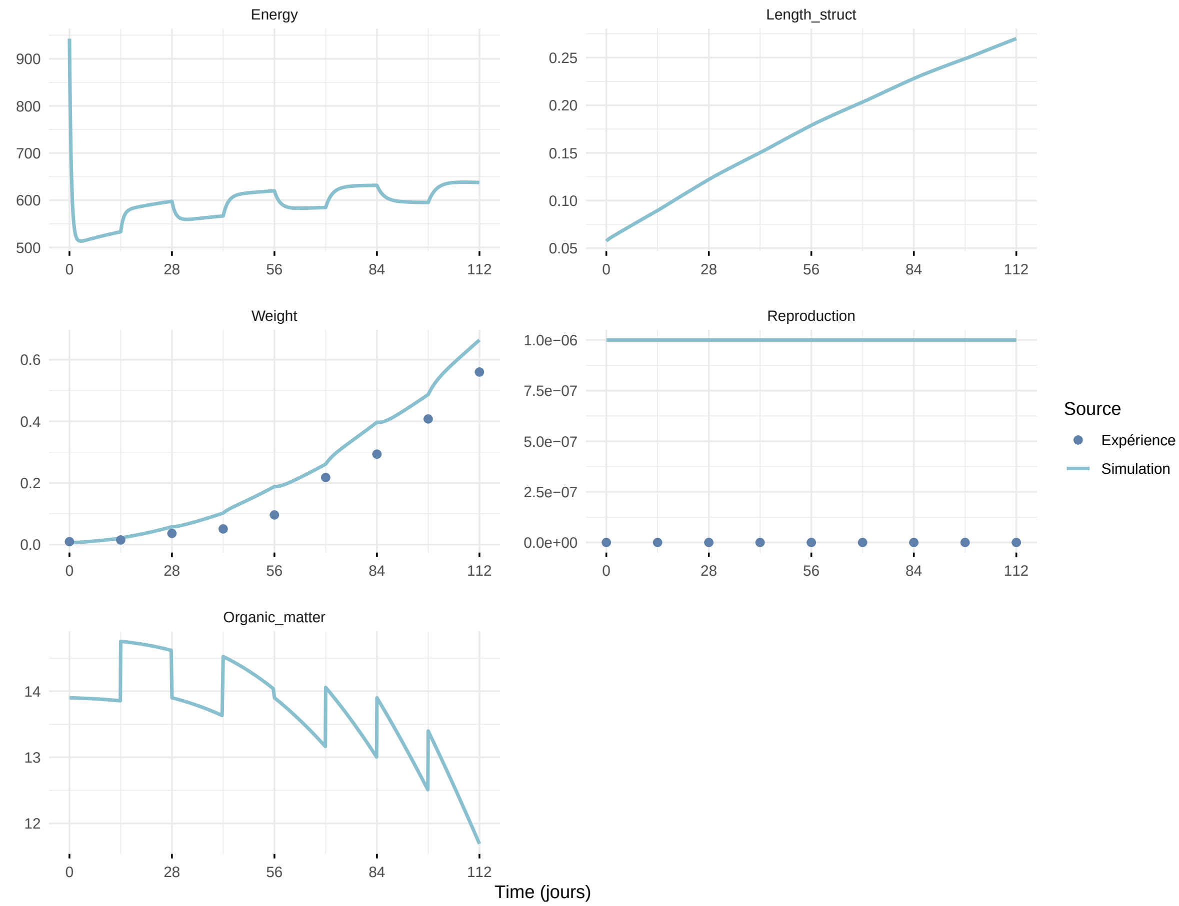
Expérience : Bart2019_XP1_1
Energy
Length_struct
Weight
Reproduction
Organic_matter
Source
Expérience
Simulation
Time (jours)

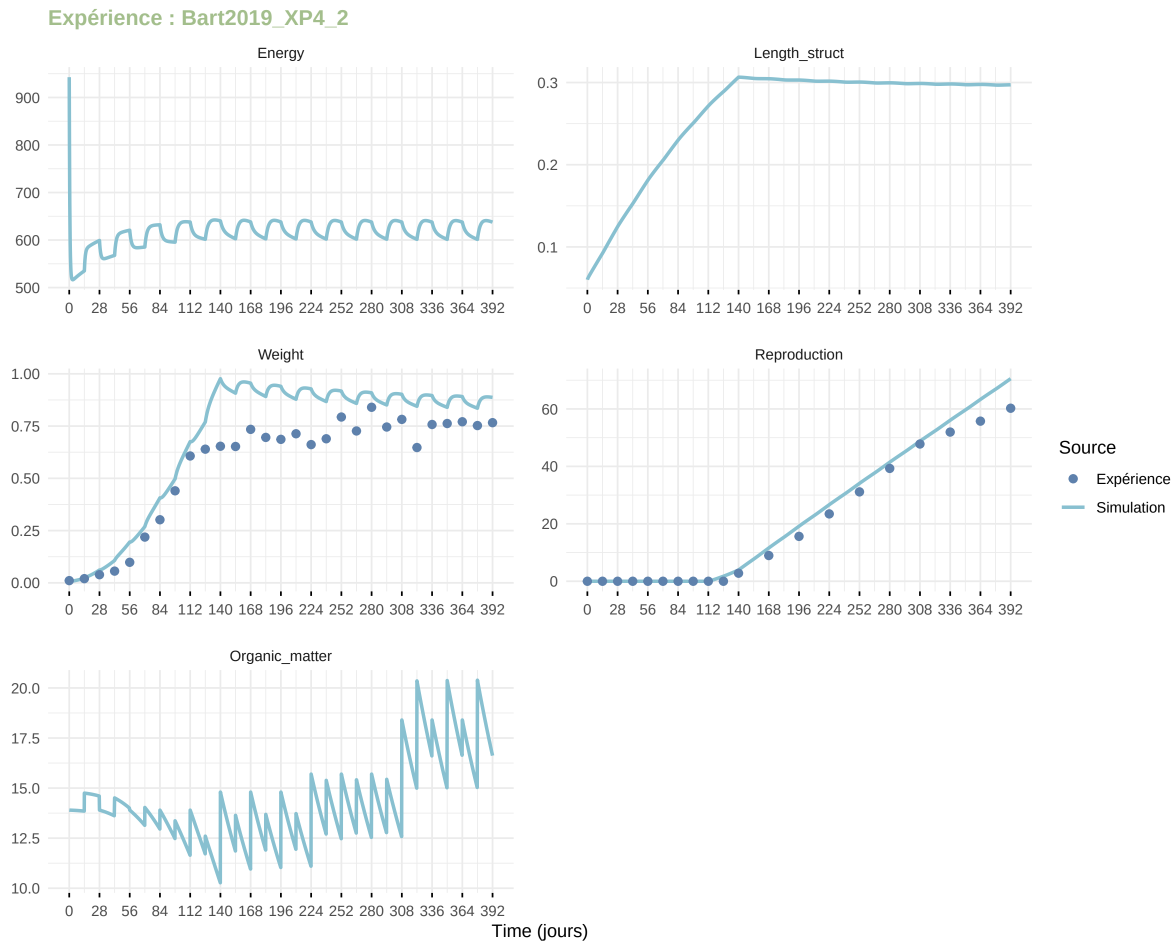
Expérience : Bart2019_XP1_2



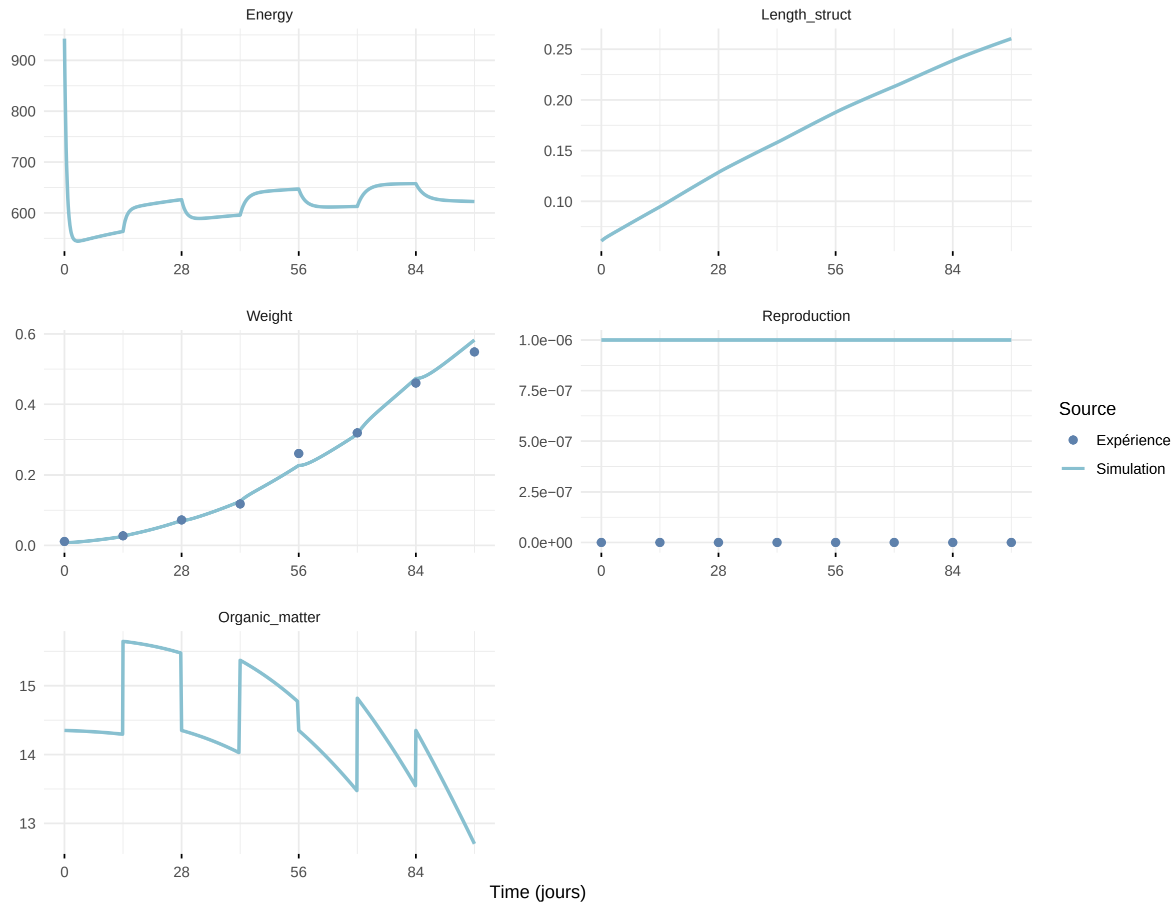

Expérience : Bart2019_XP3
Energy
Length_struct
Weight
Reproduction
Organic_matter
Source
Expérience
Simulation
Time (jours)

Expérience : Bart2019_XP4_1

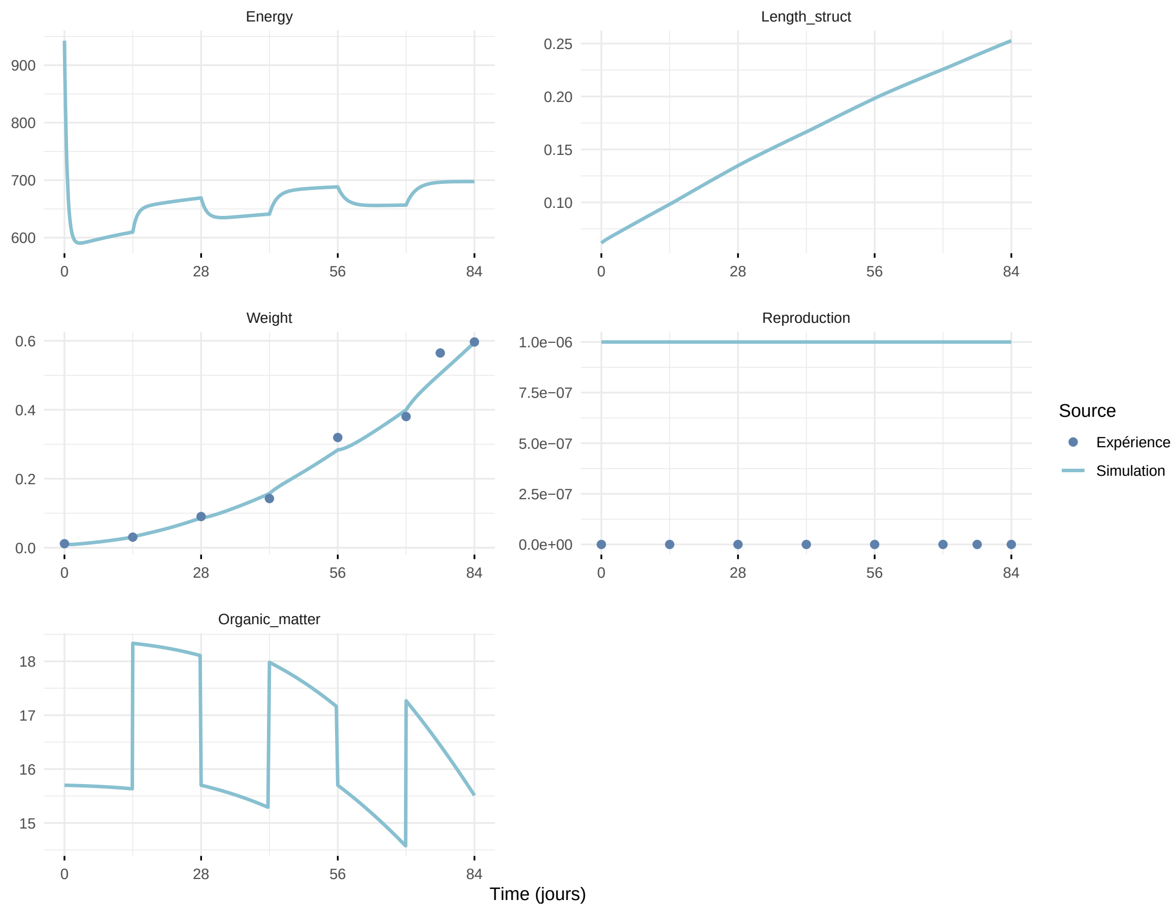


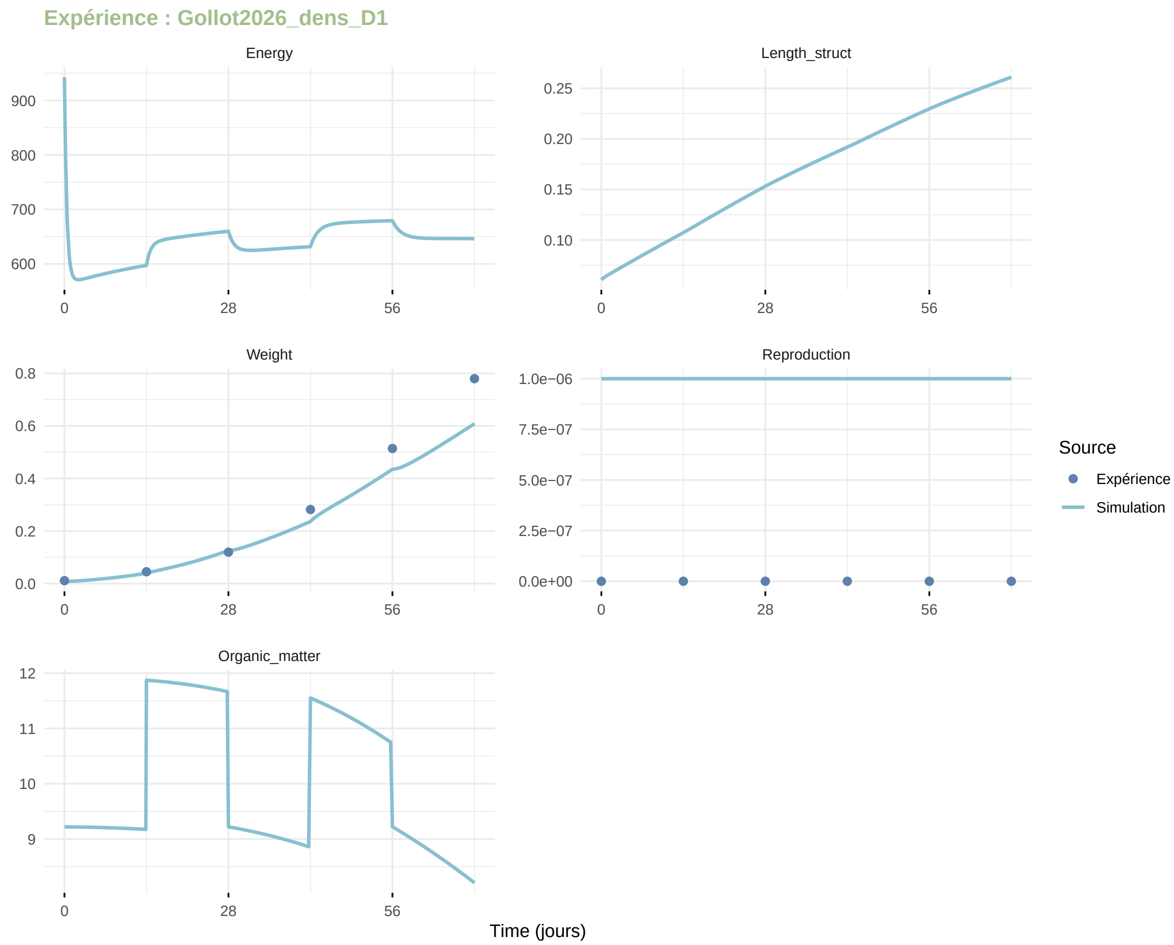

Expérience : Bart2019_XP4_2
Energy
Length_struct
Weight
Reproduction
Organic_matter
Source
Expérience
Simulation
Time (jours)

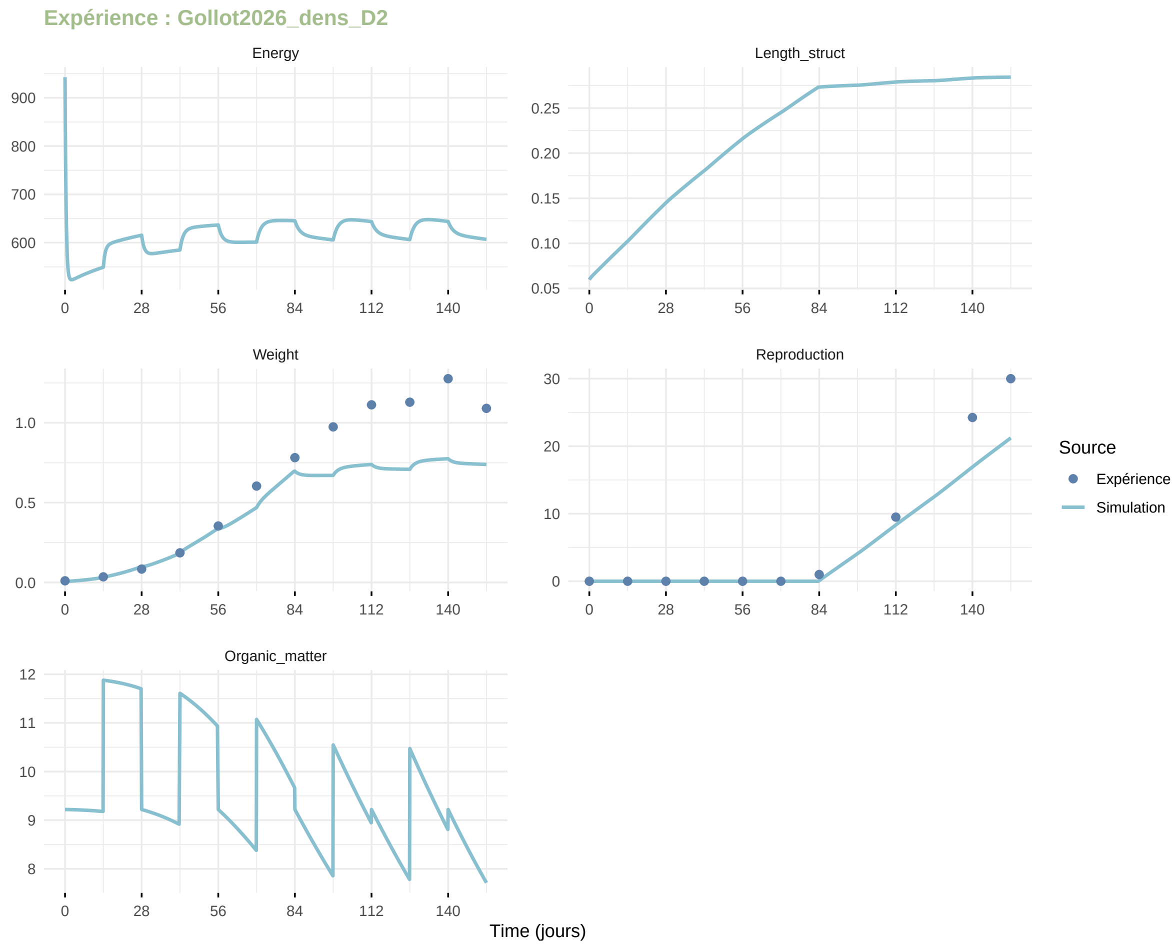
Expérience : Bart2019_XP5



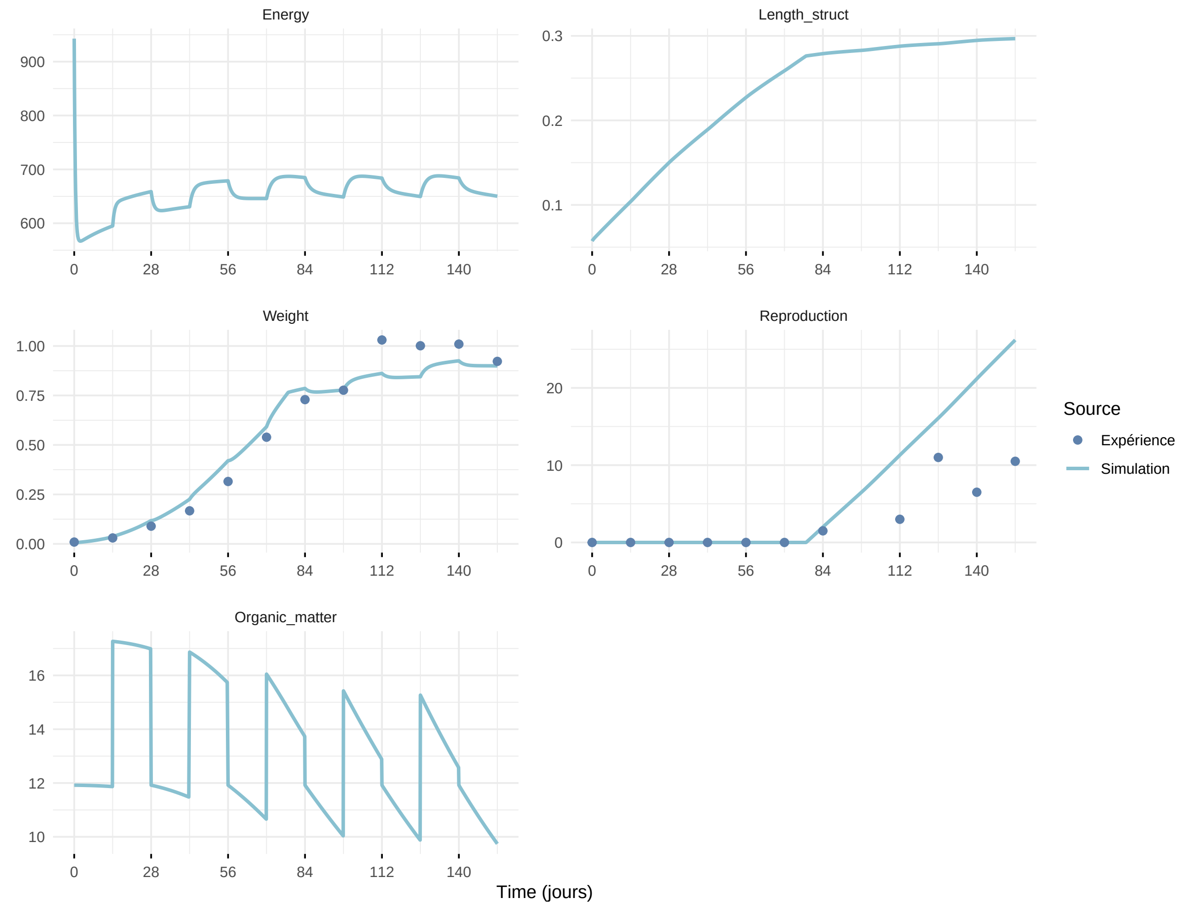
Expérience : Bart2020

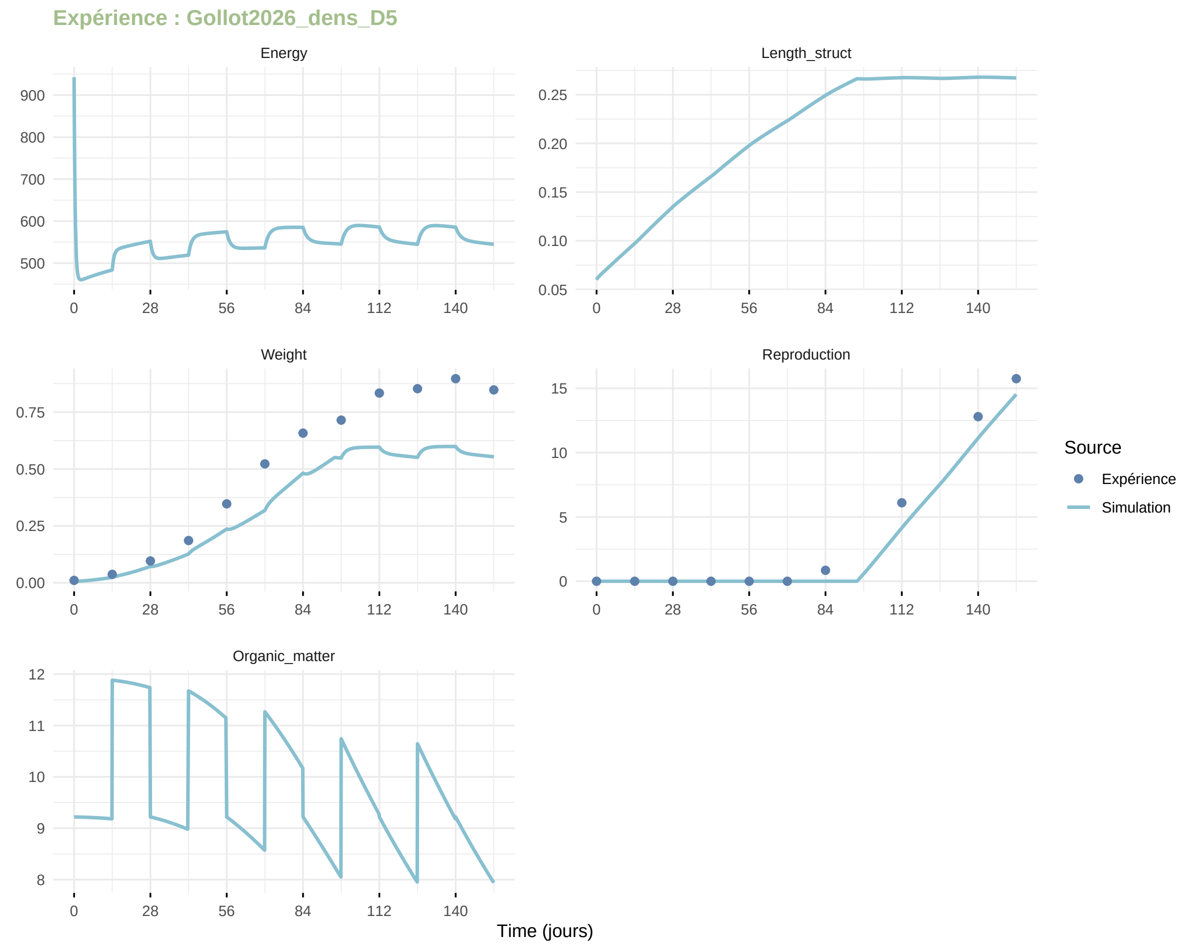



Expérience : Gollot2026_dens_D1
Energy
900
800
700
600
0
28
56
Length_struct
0.25
0.20
0.15
0.10
Weight
0.8
0.6
0.4
0.2
0.0
Reproduction
1.0e−06
7.5e−07
5.0e−07
2.5e−07
0.0e+00
Organic_matter
12
11
10
9
Source
Expérience
Simulation
Time (jours)


Expérience : Gollot2026_dens_D2
Energy
Length_struct
Weight
Reproduction
Organic_matter
Source
Expérience
Simulation
Time (jours)

Expérience : Gollot2026_dens_D2b




Expérience : Gollot2026_dens_D5
Energy
Length_struct
Weight
Reproduction
Organic_matter
Source
Expérience
Simulation
Time (jours)

Expérience : Gollot2026_dens_D7